

Scalability & Future Plan

Date	3 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited support for concurrent users.	Application performance may decrease when many users access the system simultaneously.	High
2	CSV-based data storage.	Not suitable for managing large-scale user data and long-term interaction history.	High
3	Dependency on Google Gemini API.	AI guidance requires an active internet connection and API availability.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local machine.	Deploy on Google Cloud , AWS , or Azure with auto-scaling and load balancing.
2	Data Storage	Stores interaction history in CSV files.	Replace CSV with MongoDB , PostgreSQL , or Firebase for scalable cloud-based storage.
3	AI Performance	Emotion prediction runs sequentially on local hardware.	Optimize inference using model caching, asynchronous processing, and GPU acceleration.
4	Security	Basic API key protection through .env configuration.	Implement OAuth authentication, encrypted data storage, secure APIs, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Database integration, user authentication, and cloud deployment.	2 Months	Enables secure user accounts, scalable storage, and online accessibility.
Phase 3	Enhanced AI guidance with multilingual support and personalized learning history.	4 Months	Improves user engagement, accessibility, and learning effectiveness.
Phase 4	Mobile application, advanced analytics dashboard, teacher portal, and real-time emotion monitoring.	6 Months	Expands platform accessibility, improves collaboration between students and educators, and provides comprehensive learning insights.